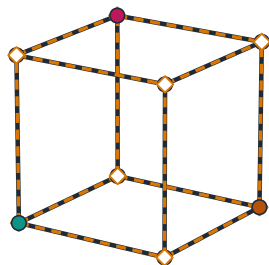


controlled factors  
same-population fit/eval

### dSprites

VICReg, ResNet-18

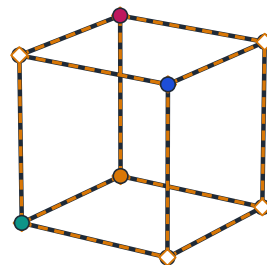


size  
x-position  
y-position

RMSE  $2.6 \times 10^{-5}$  max|cos|  $1.8 \times 10^{-5}$

### 3DShapes

VICReg, ResNet-18

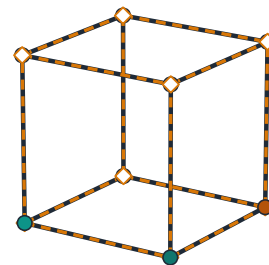


object hue  
shape  
size

RMSE  $3.8 \times 10^{-4}$  max|cos|  $8.1 \times 10^{-5}$

### dSprites

VICReg, ResNet-50

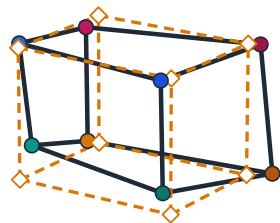


size  
x-position  
y-position

RMSE  $4.0 \times 10^{-5}$  max|cos|  $2.3 \times 10^{-5}$

### CelebA

VICReg, pretrained on CelebA

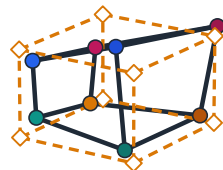


wearing lipstick  
high cheekbones  
black hair

RMSE 0.163 max|cos| 0.057

### CelebA

I-JEPA, pretrained on CelebA

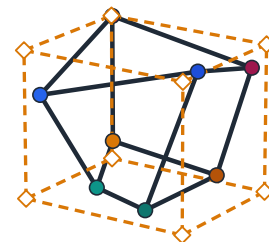


heavy makeup  
black hair  
smiling

RMSE 0.274 max|cos| 0.119  
misses fixed RMSE criterion (0.25)

### CUB-200

VICReg, pretrained on ImageNet



breast color: white  
primary color: black  
breast pattern: solid

RMSE 0.334 max|cos| 0.185

natural images  
train-fit / held-out test